Practice Problems 3: Solutions

ECON 441: Introduction to Mathematical Economics — Instructor: Div Bhagia

<u>Exercise 4.2</u>

1. $A = \begin{bmatrix} 7 & -1 \\ 6 & 9 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 4 \\ 3 & -2 \end{bmatrix} \quad C = \begin{bmatrix} 8 & 3 \\ 6 & 1 \end{bmatrix}$

   (a) $A + B = \begin{bmatrix} 7 & 3 \\ 9 & 7 \end{bmatrix}$

   (b) $C - A = \begin{bmatrix} 1 & 4 \\ 0 & -8 \end{bmatrix}$

   (c) $3A = \begin{bmatrix} 21 & -3 \\ 18 & 27 \end{bmatrix}$

   (d) $4B + 2C = \begin{bmatrix} 0 & 16 \\ 12 & -8 \end{bmatrix} + \begin{bmatrix} 16 & 6 \\ 12 & 2 \end{bmatrix} = \begin{bmatrix} 16 & 22 \\ 24 & -6 \end{bmatrix}$

2. $A = \begin{bmatrix} 2 & 8 \\ 3 & 0 \\ 5 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 0 \\ 3 & 8 \end{bmatrix} \quad C = \begin{bmatrix} 7 & 2 \\ 6 & 3 \end{bmatrix}$

   (a) $\boldsymbol{AB}$ is defined as number of columns in $\boldsymbol{A}$ is two which is equal to the number of rows in $\boldsymbol{B}$.

$$AB = \begin{bmatrix} 2\times 2 + 8\times 3 & 2\times 0 + 8\times 8 \\ 3\times 2 + 0\times 3 & 3\times 0 + 0\times 8 \\ 5\times 2 + 1\times 3 & 5\times 0 + 1\times 8 \end{bmatrix} = \begin{bmatrix} 28 & 64 \\ 6 & 0 \\ 13 & 8 \end{bmatrix}$$

Not possible to calculate BA as $\boldsymbol{B}$ has two columns, but $\boldsymbol{A}$ has three rows.

(b) $\boldsymbol{BC}$ and $\boldsymbol{CB}$ are both defined as both have two rows and two columns.

$$BC = \begin{bmatrix} 14 & 4 \\ 69 & 30 \end{bmatrix} \neq CB = \begin{bmatrix} 20 & 16 \\ 21 & 24 \end{bmatrix}$$

4.

(a) $\begin{bmatrix} 0 & 2 \\ 36 & 20 \\ 16 & 3 \end{bmatrix}_{3\times 2}$

(b) $\begin{bmatrix} 49 & 3 \\ 4 & 3 \end{bmatrix}_{2\times 2}$

(c) $\begin{bmatrix} 3x + 5y \\ 4x + 2y - 7z \end{bmatrix}_{2\times 1}$

(d) $\begin{bmatrix} 7a + c & 2b + 4c \end{bmatrix}$